Fundamentals of acoustics – sound for engineers

Acoustics is the physics behind everything we hear – and everything we would rather not hear. At DTH the course was called “Fundamentals of acoustics for mechanical engineers”, and it laid the foundation on which all the later acoustics courses built: the wave equation, the decibel scale, impedance and intensity. This page goes through the basic ideas with two interactive figures.

> DTH, course 8042 (1987), 6 points: Fundamentals of acoustics for mechanical engineers, Acoustics Laboratory. The course laid the foundation for the other acoustics courses on this list and for the thesis on sound propagation in ducts with absorbent linings.

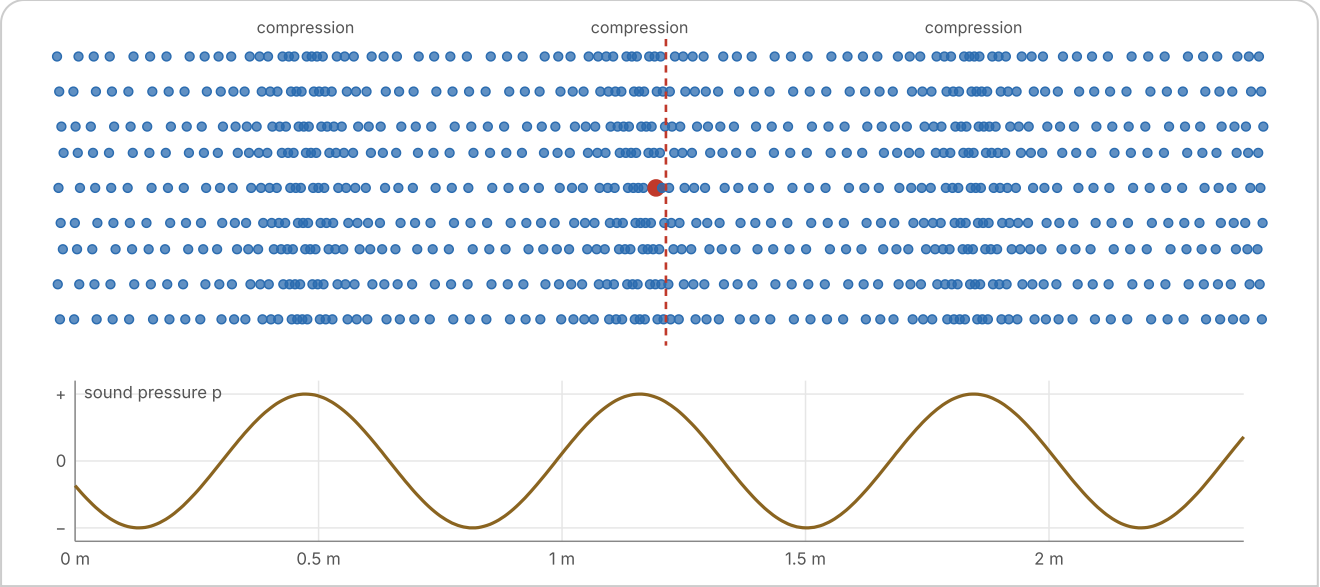
The wave equation

Sound consists of small changes in pressure and density that propagate through a medium. Atmospheric pressure is about 101 325 Pa, and even a very loud sound changes it by only a few pascals. The three basic equations – conservation of mass, Euler’s equation of motion and the adiabatic equation of state – can therefore be linearised, giving the acoustic wave equation for the sound pressure p:

$$\nabla^2 p - \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} = 0, \qquad c = \sqrt{\frac{\gamma p_0}{\rho_0}} = \sqrt{\frac{\gamma R T}{M}}$$

For air at 20 °C, $c \approx 343$ m/s. In the *Principia* Newton calculated the speed of sound from the isothermal compressibility and got about 15 % too little; only in 1816 did Laplace realise that the compressions happen so fast that the heat has no time to flow away – the process is adiabatic, and the factor $\gamma = 1.4$ enters.

The particle velocity $\vec{u}$ is related to the pressure by Euler’s equation $\rho_0 \, \partial \vec{u}/\partial t = -\nabla p$. For harmonic vibrations one writes $p = \mathrm{Re}\{\hat{p}\, e^{j\omega t}\}$, and the wave equation becomes the Helmholtz equation $\nabla^2 \hat{p} + k^2 \hat{p} = 0$ with the wavenumber $k = \omega / c$.



Sound pressure and decibels

The ear spans an enormous range: from the threshold of hearing at about 20 µPa to the threshold of pain at about 20–60 Pa – a factor of more than a million in pressure and 10^{12} in intensity. A logarithmic scale is therefore used. The sound pressure level is

$$L_p = 20 \log_{10} \frac{p_{\mathrm{rms}}}{p_0} \ \mathrm{dB}, \qquad p_0 = 20\ \mu\mathrm{Pa}$$

The decibel is not a unit but a way of writing a ratio. This has practical consequences: two equally loud, uncorrelated sources do not give twice as many decibels, only 3 dB more, because it is the energies that add: $L = 10 \log_{10} \left(10^{L_1/10} + 10^{L_2/10} \right)$. Ten identical machines give 10 dB more than one. And a source 10 dB weaker than another contributes less than half a decibel – it can safely be ignored when you are trying to reduce noise.

Sound	p_{rms}	L_p
Threshold of hearing at 1 kHz	20 µPa	0 dB
Quiet bedroom at night	0.6 mPa	30 dB
Conversation at 1 m	20 mPa	60 dB
Heavy traffic at 10 m	0.6 Pa	90 dB
Pneumatic drill at 1 m	6 Pa	110 dB
Threshold of pain	20–60 Pa	120–130 dB

Impedance and intensity

In a plane wave pressure and particle velocity are in phase, and their ratio is the characteristic impedance of the medium, $Z_0 = \rho_0 c \approx 415$ Pa·s/m for air (in rayl). For water $\rho c \approx 1.5 \cdot 10^6$ rayl – 3600 times more. That is why sound hardly passes from air into water: at an interface the pressure is reflected with the coefficient $R = (Z_2 - Z_1)/(Z_2 + Z_1)$, and only 0.1 % of the energy gets through.

Sound intensity is the flow of energy per unit area, the time average of $p\,\vec{u}$. For a plane wave

$$I = \frac{p_{\mathrm{rms}}^2}{\rho_0 c}, \qquad L_I = 10 \log_{10} \frac{I}{10^{-12}\ \mathrm{W/m^2}}$$

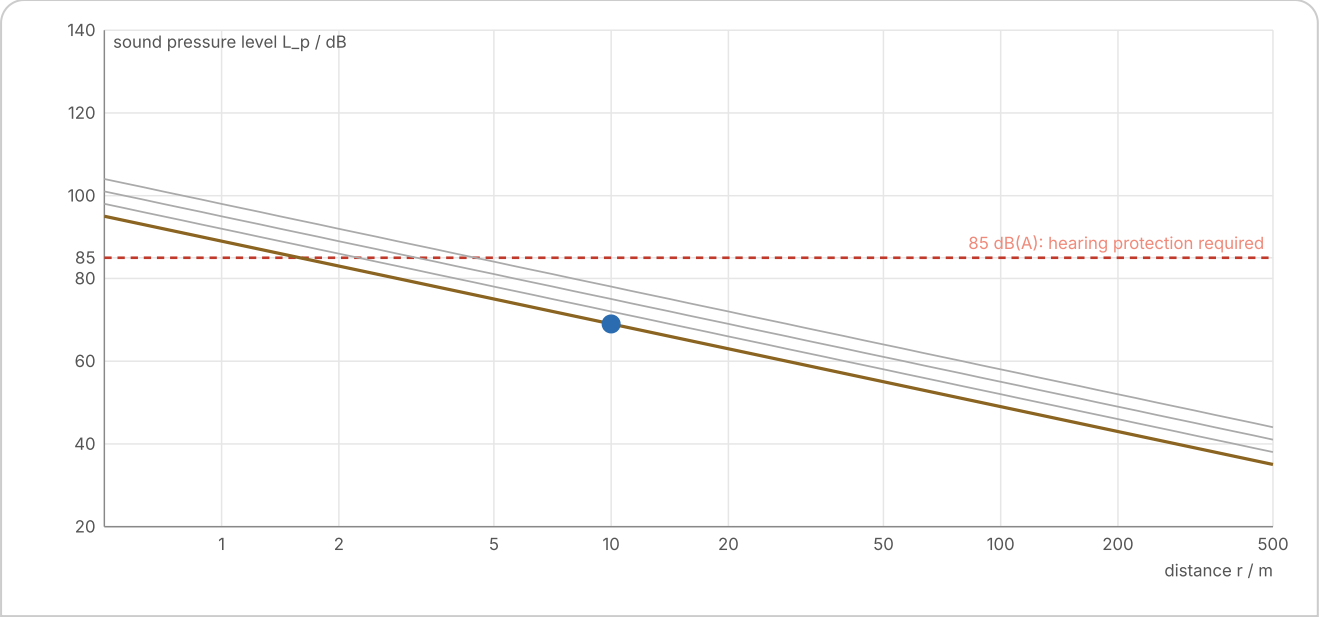
The sound power W is the intensity integrated over a surface around the source, and it is a property of the source alone – independent of the room and the distance. That is why the noise of machines is stated as a sound power level $L_W = 10 \log_{10}(W/10^{-12}\ \mathrm{W})$. A conversation has a sound power of about 10 µW, a symphony orchestra perhaps 1 W, and a jet engine more than 10 kW. Today intensity can be measured directly with two closely spaced microphones – a method developed at DTH and Brüel & Kjær around 1980 (see From Bell's inequality to sound intensity).

The point source

A small source radiating equally in all directions sends out a spherical wave: $\hat{p} = \frac{A}{r} e^{-jkr}$. The power spreads over a spherical surface $4\pi r^2$, so the intensity falls as $1/r^2$ – 6 dB for every doubling of distance. If the source stands on a hard floor it radiates into a hemisphere only, and the sound pressure is 3 dB higher; in a corner 9 dB. This is described by the directivity factor Q:

$$L_p = L_W - 20 \log_{10} \frac{r}{1\ \mathrm{m}} - 11 + 10 \log_{10} Q \ \ \mathrm{dB}$$

Close to the source (within a wavelength or the size of the source) one is in the near field, where pressure and velocity are not in phase and the formula does not hold. Indoors the diffuse field from the reflections must be added – see the page on machinery noise.



The ear and A-weighting

The ear is not equally sensitive at all frequencies. It is most sensitive between 2 and 5 kHz – the ear canal is a tube closed at one end, with its quarter-wave resonance around 3 kHz – and much less sensitive in the bass. Fletcher and Munson measured the curves of equal loudness in 1933; today they are standardised in ISO 226. For measurements a filter imitating the ear at low levels is used: *A-weighting*. An A-weighted level is written dB(A) or L_A. At 100 Hz the filter subtracts 19 dB, at 1 kHz nothing. Almost all noise limits – at work, at homes and for machines – are stated in dB(A).

Because noise varies in time, the equivalent level L_{eq} is used: the constant level that has the same energy as the varying noise over a period T,

$$L_{\mathrm{eq},T} = 10 \log_{10} \left(\frac{1}{T} \int_0^T \frac{p_A^2(t)}{p_0^2} \, dt \right)$$

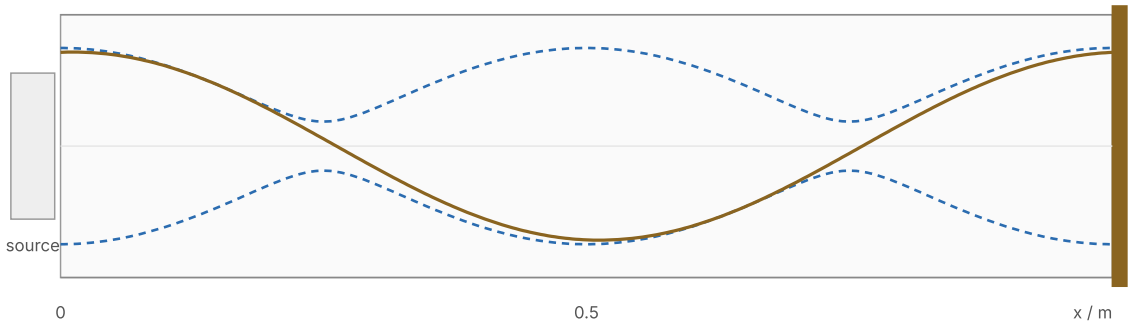
This idea – equal energy, equal damage – is the basis of noise dose at work: 85 dB(A) for 8 hours corresponds to 88 dB(A) for 4 hours and 94 dB(A) for 1 hour.

Standing waves in tubes

When a plane wave is sent into a tube, it is reflected from the end. The outgoing and reflected waves interfere and form a standing-wave pattern. At a closed, rigid end the particle velocity is zero and the pressure is maximal; at an open end the pressure is almost zero (strictly a small end correction of $0.6\,a$ further out, where a is the radius). The resonances are

$$\text{closed–closed and open–open: } f_n = \frac{nc}{2L}, \qquad \text{closed–open: } f_n = \frac{(2n-1)c}{4L}$$

That is the principle of organ pipes, the clarinet (closed–open, hence only odd harmonics) and the flute (open–open). In the laboratory a *standing-wave tube* (Kundt's tube) is used to measure the absorption coefficient of a material: one measures the ratio of maximum to minimum in the standing-wave pattern in front of the material and calculates the reflection coefficient.



Absorption in air

Over long distances sound is also attenuated by the air itself. The classical absorption from viscosity and heat conduction (Stokes and Kirchhoff) is small. The most important mechanism is *molecular relaxation*: the vibrations of oxygen and nitrogen molecules cannot keep up with the pressure changes, and energy is lost. Water vapour speeds up the relaxation, so the absorption depends strongly on humidity. At 20 °C and 50 % humidity it is about 0.1 dB/100 m at 125 Hz but 10 dB/100 m at 8 kHz (ISO 9613-1). That is why distant thunder is a deep rumble and a distant orchestra has no treble.

Since 1989

- **Computation:** Where in the 1980s one calculated with formulas and nomograms, sound fields are now simulated with finite element and boundary element methods (FEM/BEM) on an ordinary computer.
- **Instruments:** The sound level meter with a needle has become a digital analyser with real-time frequency analysis, and MEMS microphones in mobile phones are good enough for many indicative measurements.
- **Acoustic holography and microphone arrays:** With many microphones one can build "acoustic cameras" that show where noise comes from – a technique that Ørsted DTU and

Brüel & Kjær helped develop.

- **Active noise control:** The idea of cancelling noise with anti-phase dates from 1936 (Paul Lueg), but only fast signal processors made it practical – today in headphones and cars.

Further reading

- Finn Jacobsen et al.: *Fundamentals of Acoustics and Noise Control* (DTU lecture notes, many editions)
- Lawrence E. Kinsler et al.: *Fundamentals of Acoustics* (4th ed., Wiley 2000)
- Allan D. Pierce: *Acoustics. An Introduction to Its Physical Principles and Applications* (3rd ed., Springer 2019)

- On this site: Sound in ducts with flow · Sound analyser · The singing wine glass

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